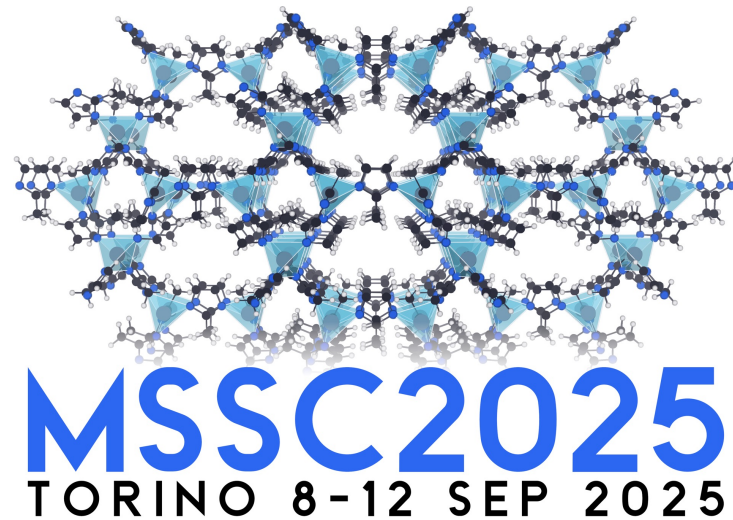


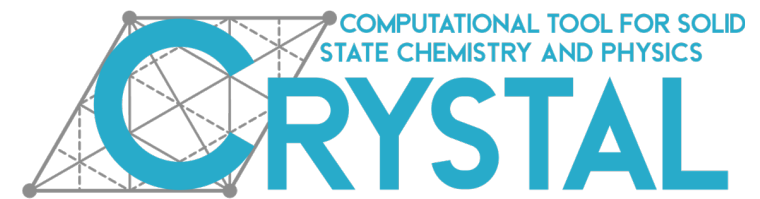


Anharmonic Vibrational States and Spectra

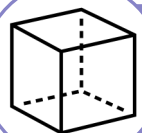
Davide Mitoli
(davide.mitoli@unito.it)



UNIVERSITÀ
DI TORINO



Outline



Limits of the HA



Anharmonic PES



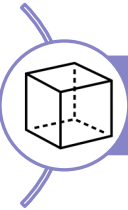
Solving the NSE



Intensities

«Quantum Chemistry is mainly about solving the electron problem»

(Someone not interested in Lattice Dynamics)

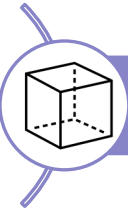


Constant-Volume Description

- ✓ Vanishing thermal expansion
- ✓ $C_P = C_V$
- ✓ Mechanical response independent from temperature

- ✓ Introduces of volume-dependent phonon frequencies
- ✓ Requires many simulations at increasing volume
- ✓ Based on the hypothesis that HA stands for each explored volume
- ✓ Provides great results in combination with DFT

Quasi-Harmonic Approximation (QHA)



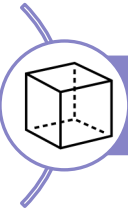
Constant Volume Approximation

- ✓ Vanishing thermal expansion
- ✓ $C_P = C_V$
- ✓ Mechanical stability from temperature

Covered by Alessandro
in the previous talk

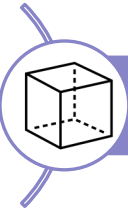
Lucky me 🙏

Quasi-Harmonic Approximation (QHA)



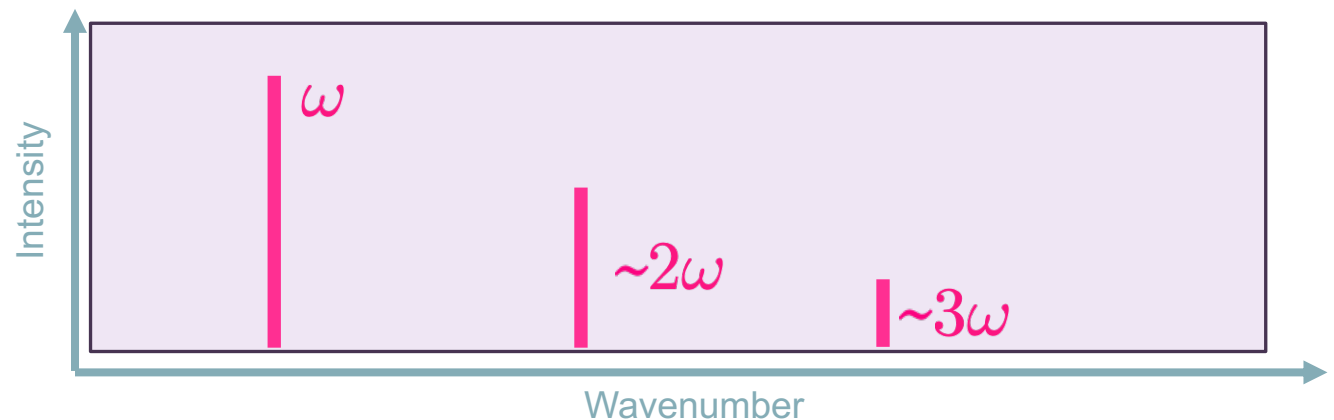
Lack of High-Order Terms

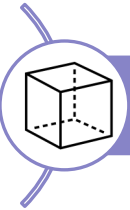
- Infinite phonon lifetime
- Infinite lattice thermal conductivity
- Temperature-independent vibrational spectra
- Lack of some spectral features



Lack of High-Order Terms

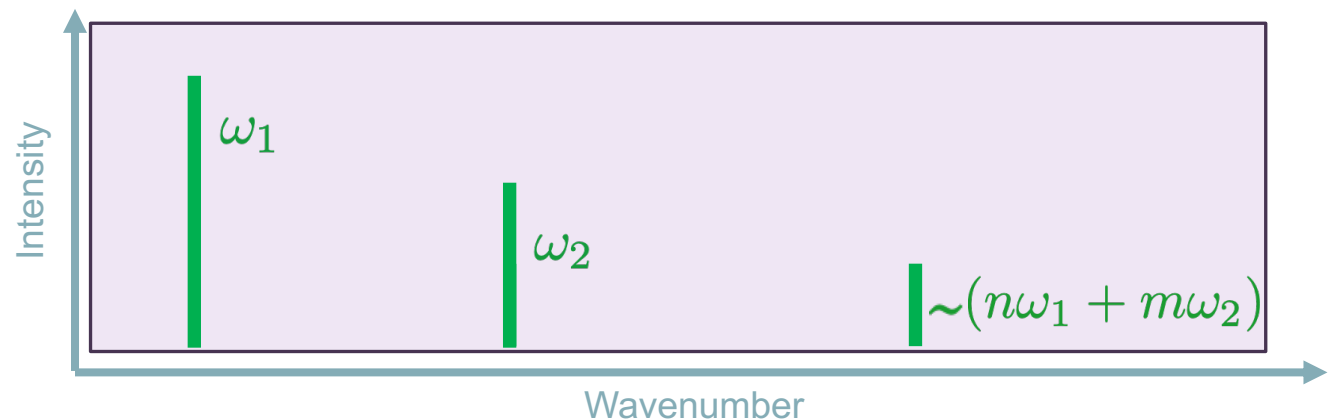
- Infinite phonon lifetime
- Infinite lattice thermal conductivity
- Temperature-independent vibrational spectra
- Lack of some spectral features:
 - Overtones

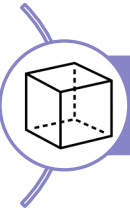




Lack of High-Order Terms

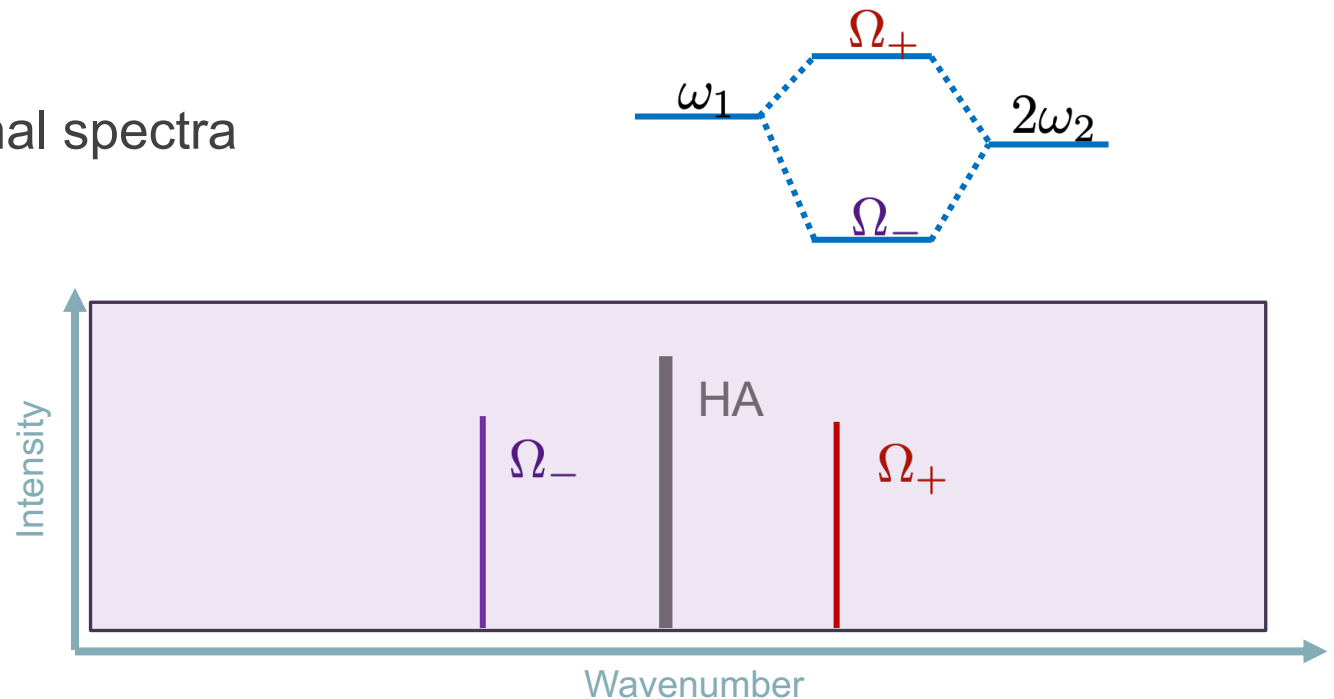
- Infinite phonon lifetime
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- Temperature-independent vibrational spectra
- Lack of some spectral features:
 - Overtones
 - Combination bands

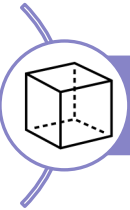




Lack of High-Order Terms

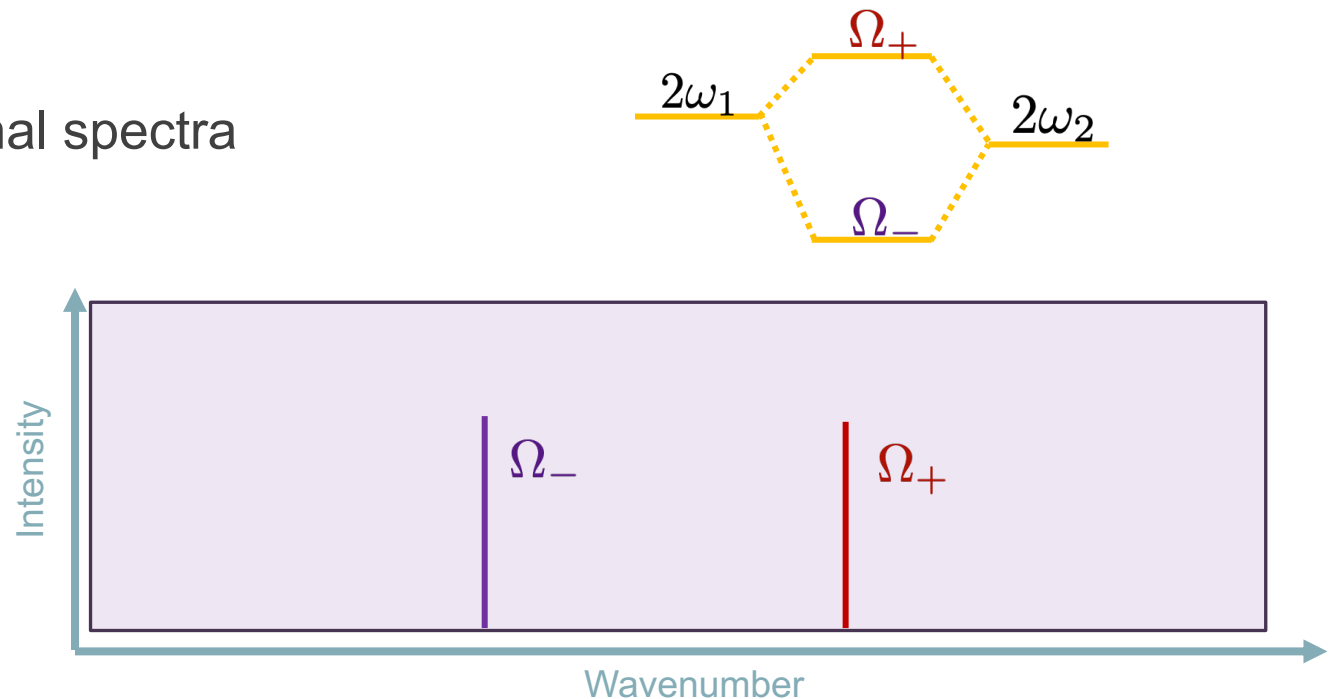
- Infinite phonon lifetime
- Infinite lattice thermal conductivity
- Temperature-independent vibrational spectra
- Lack of some spectral features:
 - Overtones
 - Combination bands
 - Fermi resonances

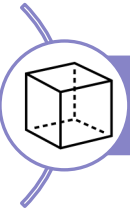




Lack of High-Order Terms

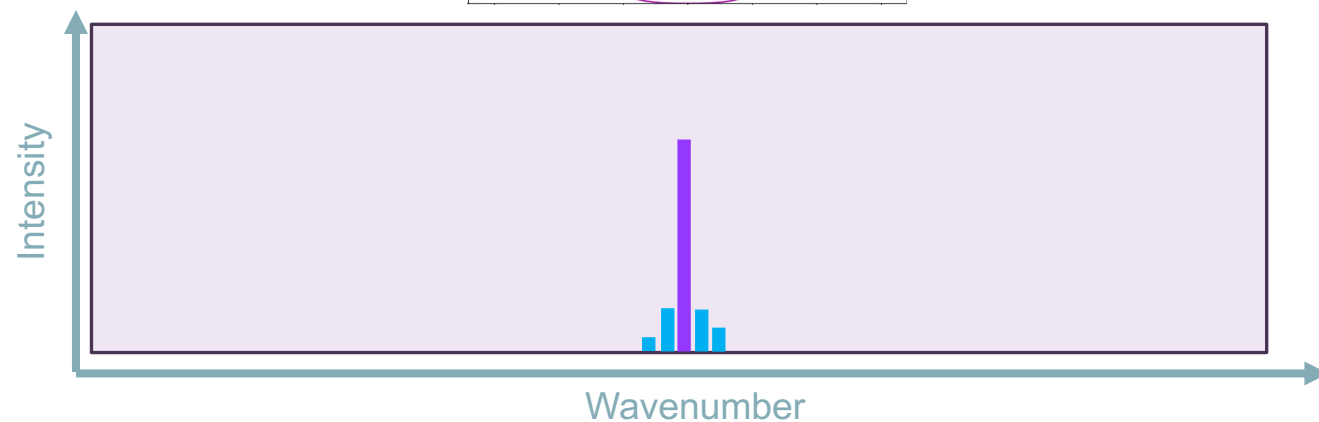
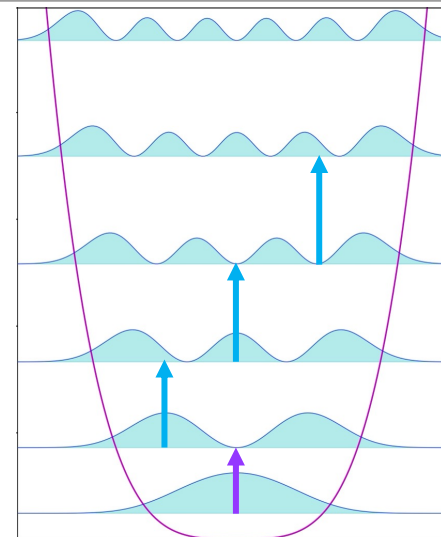
- Infinite phonon lifetime
- Infinite lattice thermal conductivity
- Temperature-independent vibrational spectra
- Lack of some spectral features:
 - Overtones
 - Combination bands
 - Fermi resonances
 - Darling-Dennison resonances





Lack of High-Order Terms

- Infinite phonon lifetime
- Infinite lattice thermal conductivity
- Temperature-independent vibrational spectra
- Lack of some spectral features:
 - Overtones
 - Combination bands
 - Fermi resonances
 - Darling-Dennison resonances
 - Hot bands





Taylor Expansion of the PES

$$E(\{Q\}) = \underbrace{\frac{1}{2} \sum_{i=1}^M \omega_i^2 Q_i^2}_{\text{Harmonic PES}} + \underbrace{\frac{1}{3!} \sum_{i,j,k=1}^M \eta_{ijk} Q_i Q_j Q_k + \frac{1}{4!} \sum_{i,j,k,l=1}^M \eta_{ijkl} Q_i Q_j Q_k Q_l + \dots}_{\text{HOTs}}$$



Taylor Expansion of the PES

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where:

ω_i is the harmonic frequency,



Taylor Expansion of the PES

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Taylor Expansion of the PES

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where:

ω_i is the harmonic frequency.

Q_i is the vibrational normal mode,

$$\eta_{ijk} = \left. \frac{\partial^3 E}{\partial Q_i \partial Q_j \partial Q_k} \right|_{\text{eq}}$$



Taylor Expansion of the PES

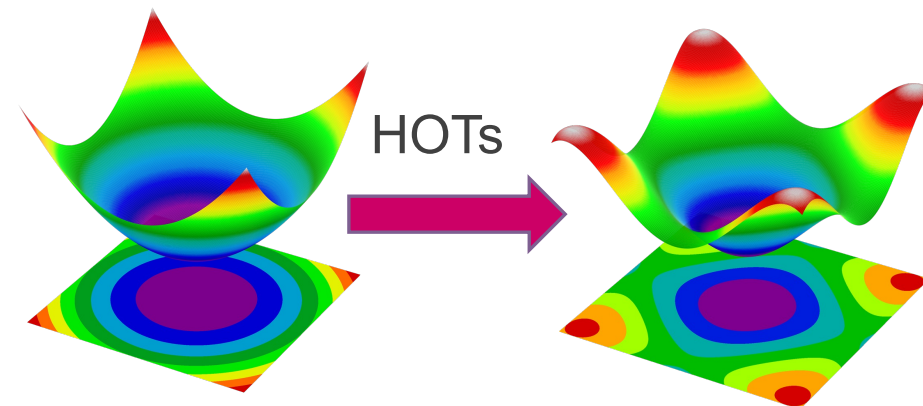
$$E(\{Q\}) = \frac{1}{2} \sum_{i=1}^M \omega_i^2 Q_i^2 + \frac{1}{3!} \sum_{i,j,k=1}^M \eta_{ijk} Q_i Q_j Q_k + \frac{1}{4!} \sum_{i,j,k,l=1}^M \eta_{ijkl} Q_i Q_j Q_k Q_l + \dots$$

where:

ω_i is the harmonic frequency.

Q_i is the vibrational normal mode.

$$\left. \begin{aligned} \eta_{ijk} &= \left. \frac{\partial^3 E}{\partial Q_i \partial Q_j \partial Q_k} \right|_{\text{eq}} \\ \eta_{ijkl} &= \left. \frac{\partial^4 E}{\partial Q_i \partial Q_j \partial Q_k \partial Q_l} \right|_{\text{eq}} \end{aligned} \right\} \text{anharmonic force constants}$$





Taylor Expansion of the PES

$$E(\{Q\}) = \frac{1}{2} \sum_{i=1}^M \omega_i^2 Q_i^2 + \frac{1}{3!} \sum_{i,j,k=1}^M \eta_{ijk} Q_i Q_j Q_k + \frac{1}{4!} \sum_{i,j,k,l=1}^M \eta_{ijkl} Q_i Q_j Q_k Q_l + \dots$$

- ✓ All derivatives beyond 1st order are computed numerically
- ✓ The number of derivatives increases as a function of the order (!!!)
- ✓ We must look for a compromise between accuracy and cost



Taylor Expansion of the PES

$$E(\{Q\}) = \frac{1}{2} \sum_{i=1}^M \omega_i^2 Q_i^2 + \frac{1}{3!} \sum_{i,j,k=1}^M \eta_{ijk} Q_i Q_j Q_k + \frac{1}{4!} \sum_{i,j,k,l=1}^M \eta_{ijkl} Q_i Q_j Q_k Q_l + \dots$$

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«*nMmT*» representation of the PES

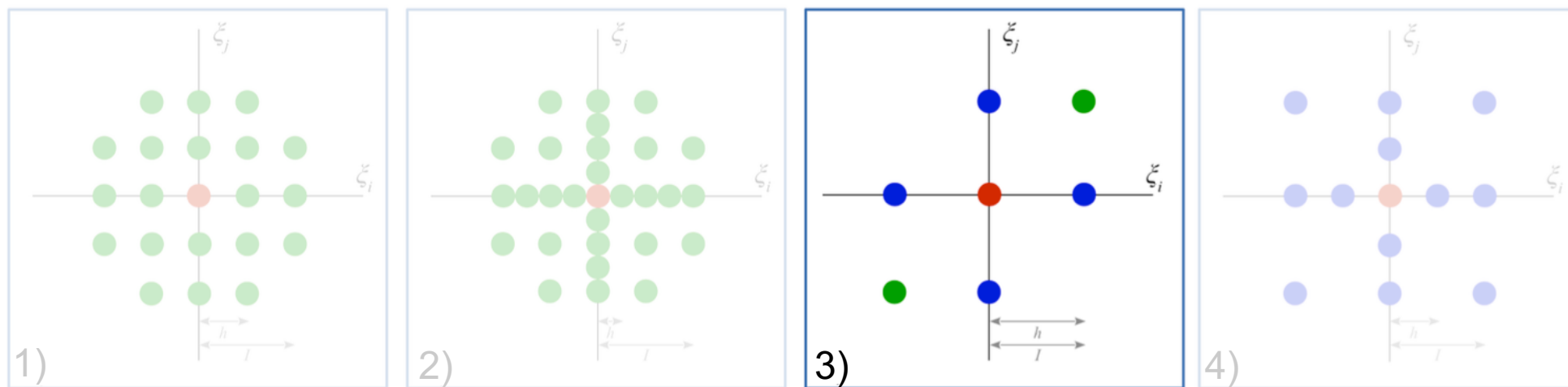
n^{th} order energy derivatives

coupling between m modes



Evaluation of HOTs: EGH

- ✓ 4 numerical strategies have been implemented so far in CRYSTAL
- ✓ EGH schemes involve energy, gradient and Hessian
- ✓ For each pair of modes i, j we need grids describing nuclear configurations

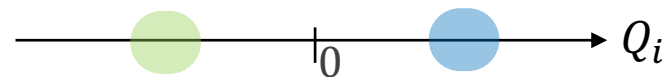




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$$\eta_{iii} = \frac{1}{s_i^2} (G_{-1}^i + G_1^i)$$

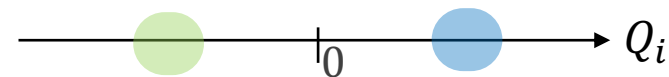




Evaluation of HOTs: EGH

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$$\eta_{ij} = \frac{1}{s_i^2} (G_{-1,0}^j + G_{1,0}^j)$$

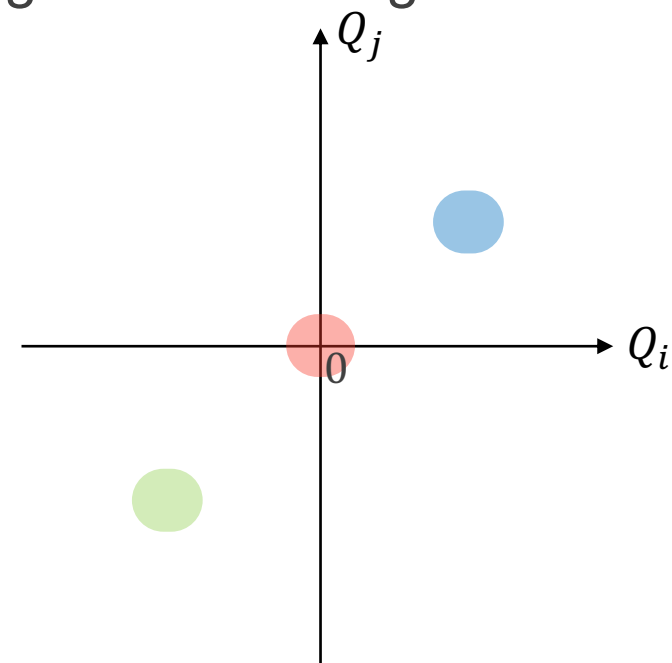




Evaluation of HOTs: EGH

- ✓ 4 numerical strategies have been implemented so far in CRYSTAL
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$$\eta_{iijj} = -\frac{1}{2s_i^2 s_j^2} (8E_{0,0} - 4E_{-1,-1} - 4E_{1,1} - s_j G_{0,-1}^j + s_j G_{0,1}^j + s_i G_{-1,0}^i + s_i G_{1,0}^i + 4s_j G_{-1,0}^j + 4s_j G_{1,0}^j + 4s_i G_{0,-1}^i + 4s_i G_{0,1}^i + 2s_i^2 \omega_i + 2s_j^2 \omega_j)$$

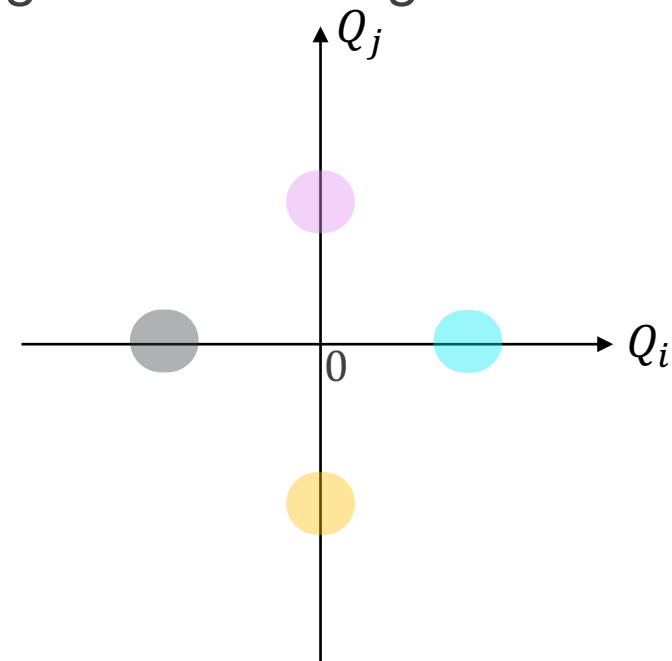




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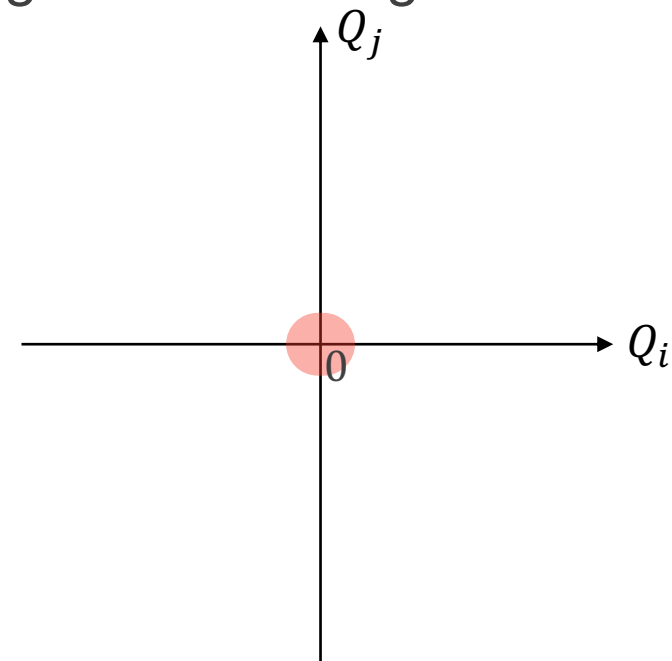




Evaluation of HOTs: EGH

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Evaluation of HOTs: EGH

- ✓ 4 numerical strategies
- ✓ EGH schemes involve
- ✓ For each pair of modes

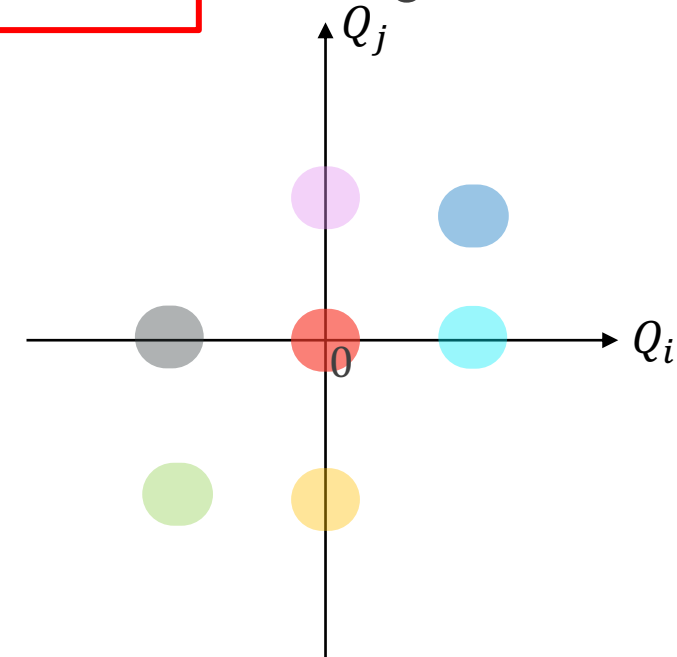
We need this procedure for each couple of modes i, j

Extremely high cost!

CRYSTAL

linear configurations

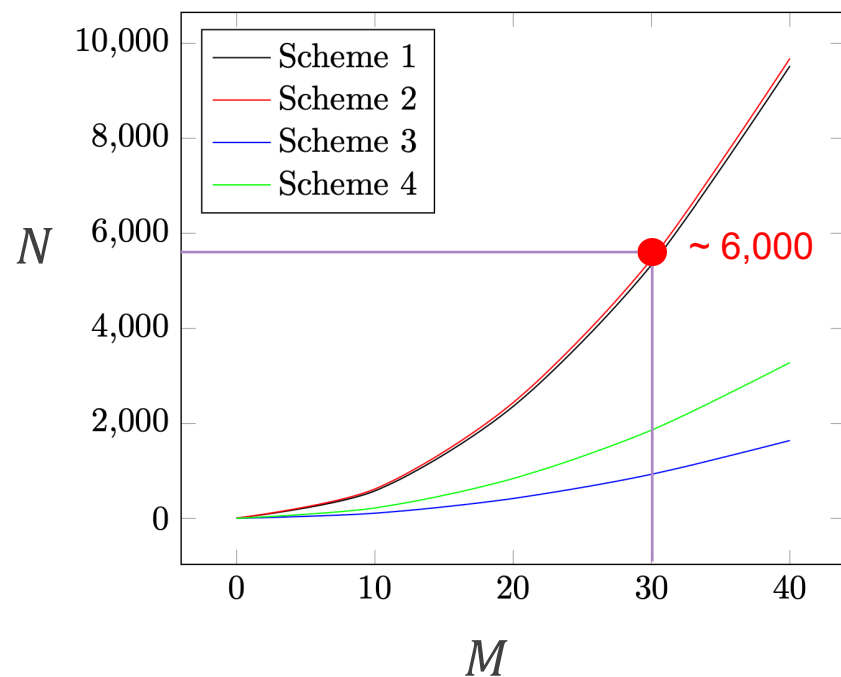
$$\begin{aligned} \eta_{iijj} = & -\frac{1}{2s_i^2 s_j^2} (8E_{0,0} - 4E_{-1,-1} - 4E_{1,1} - s_j G_{0,-1}^j + s_j G_{0,1}^j \\ & + s_i G_{-1,0}^i + s_i G_{1,0}^i + 4s_j G_{-1,0}^j + 4s_j G_{1,0}^j + 4s_i G_{0,-1}^i \\ & + 4s_i G_{0,1}^i + 2s_i^2 \omega_i + 2s_j^2 \omega_j) \end{aligned}$$





Evaluation of HOTs: Performance

- ✓ The number N of nuclear configurations (a.k.a. SCF cycles + gradients) to be explicitly explored depends on the specific numerical scheme

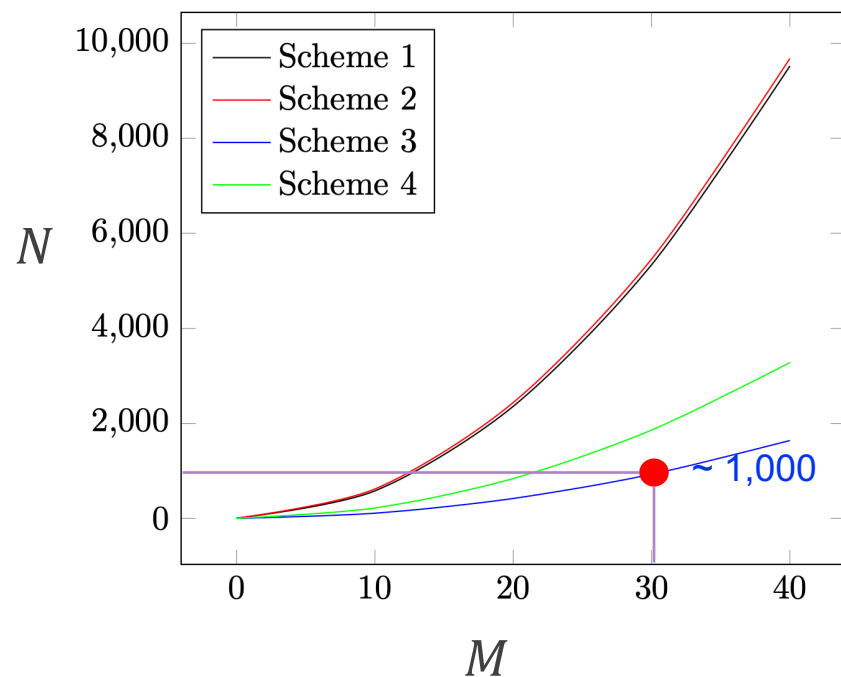


$$N_{\text{scheme 2}} = 1 + 8M + 12 \binom{M}{2}$$



Evaluation of HOTs: Performance

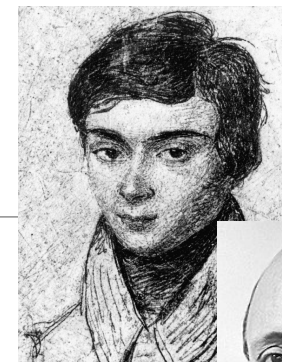
- ✓ The number N of nuclear configurations (a.k.a. SCF cycles + gradients) to be explicitly explored depends on the specific numerical scheme



$$N_{\text{scheme 3}} = 1 + 2M + 2 \binom{M}{2}$$

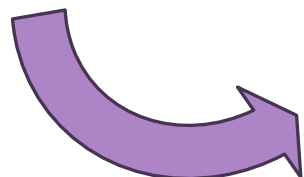


Exploitation of Point Symmetry

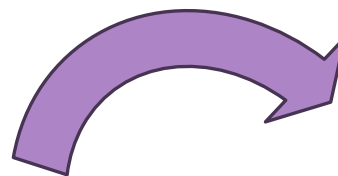


- ✓ Do we need to explicitly evaluate each anharmonic force constant?
- ✓ Group theory shows that sets of HOTs may be linearly dependent!

$$\hat{P}^\alpha = \frac{1}{|\mathcal{P}|} \sum_{r=1}^{|\mathcal{P}|} \chi^\alpha(\hat{R}_r)^* \hat{R}_r$$



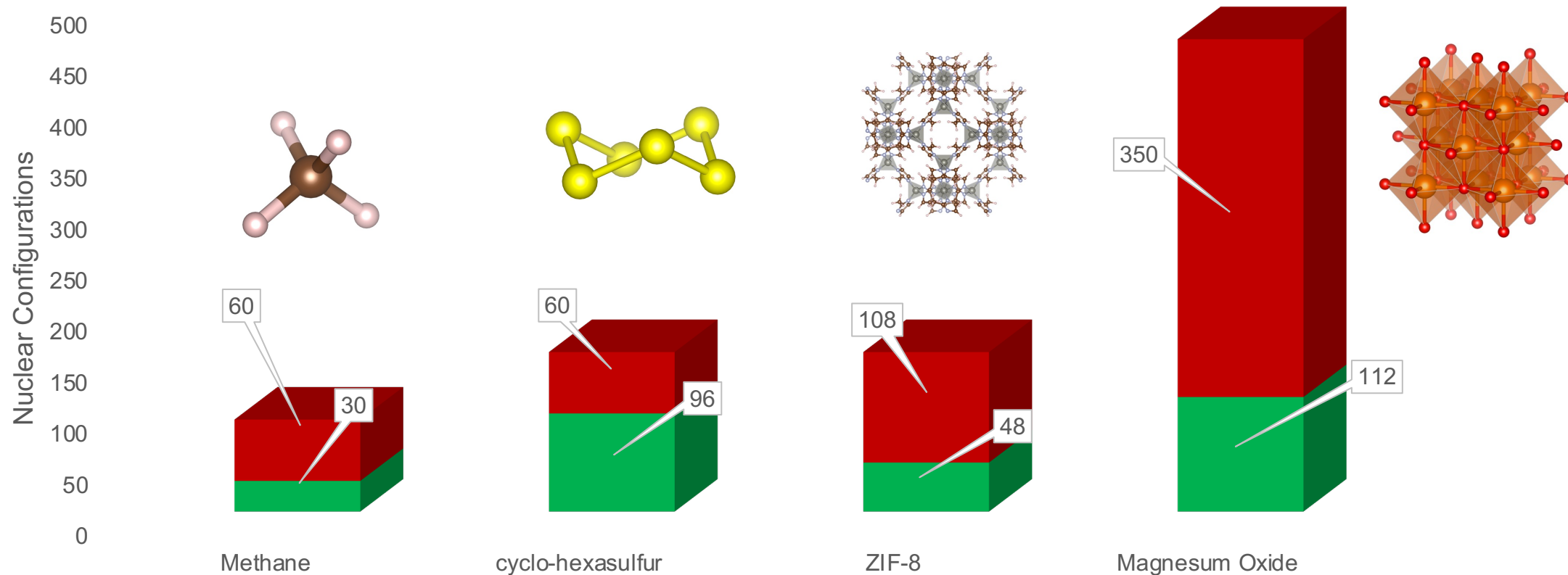
$$E(\{Q\})$$



$$\begin{cases} \sum_{t_s}^{n_s} C'_{1,t_s} |t_s\rangle = - \sum_{t_{ec}}^{n_{ec}} C'_{1,t_{ec}} |t_{ec}\rangle \\ \sum_{t_s}^{n_s} C'_{2,t_s} |t_s\rangle = - \sum_{t_{ec}}^{n_{ec}} C'_{2,t_{ec}} |t_{ec}\rangle \\ \vdots \\ \sum_{t_s}^{n_s} C'_{n_{rt},t_s} |t_s\rangle = - \sum_{t_{ec}}^{n_{ec}} C'_{n_{rt},t_{ec}} |t_{ec}\rangle \end{cases}$$

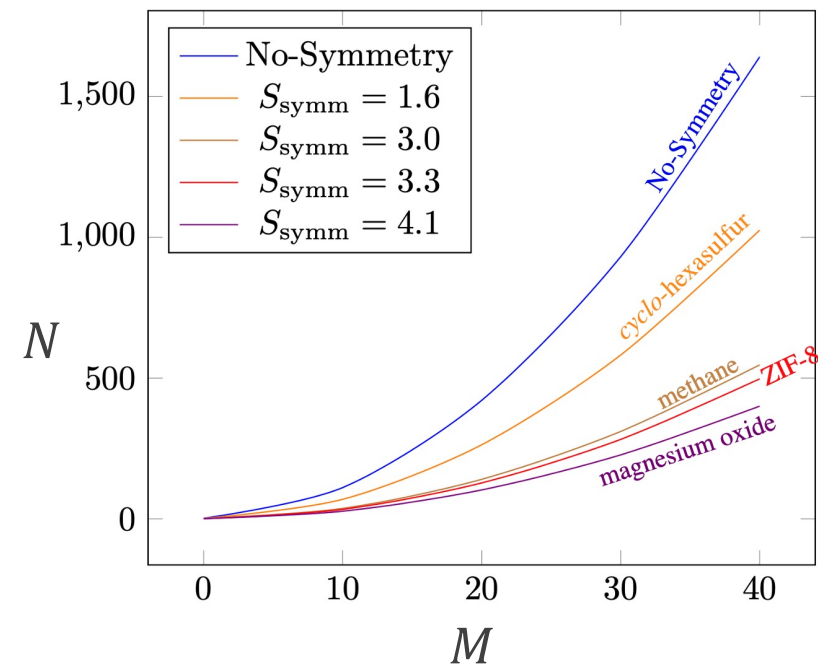
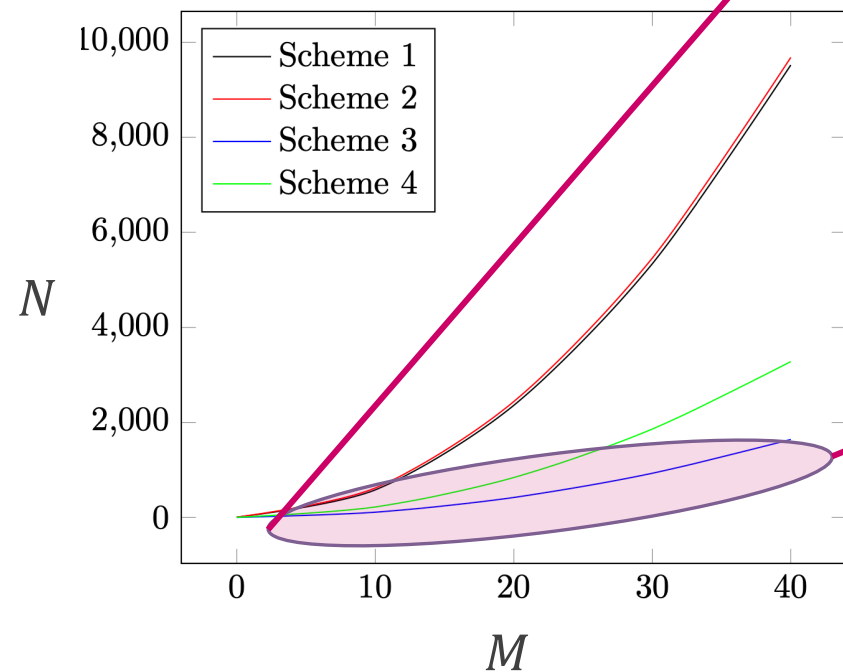


Exploitation of Point Symmetry





Exploitation of Point



$$S_{\text{symm}} = \frac{\text{nuclear confs in no-symm calculation}}{\text{nuclear confs in symm calculation}}$$



Vibrational WF Methods





Vibrational WF Methods



Vibrational Self-Consistent Field

- ✓ Mean-field approach
- ✓ Ansatz: $\Phi^n(\{Q\}) = \prod_{i=1}^M \phi_i^{n_i}(Q_i) + \text{variational}$
- ✓ Iterative approach
- ✓ Cost-effective



Vibrational WF Methods



Vibrational Configuration Interaction

- ✓ Ansatz: $\Psi_s(\{Q\}) = \sum_{n=1}^{N_{\text{conf}}} A_{n,s} \Phi^n(\{Q\})$
- ✓ In principle «exact»
- ✓ Expansion truncated according to 2 criteria:
 - ✓ n. quanta of excitation
 - ✓ n. modes
- ✓ Extremely expensive



Vibrational WF Methods



Vibrational Configuration Interaction

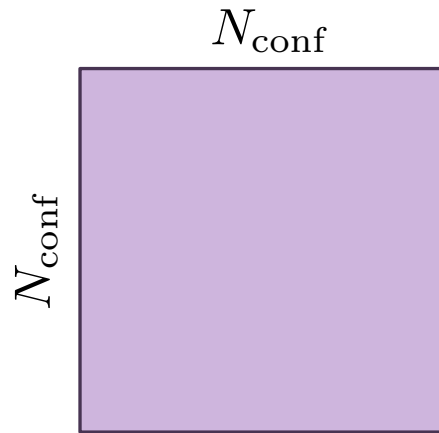
✓ Ansatz: $\Psi_s(\{Q\}) = \sum^{N_{\text{conf}}} A_{n,s} \Phi^n(\{Q\})$

Example: 4 modes, 3 quanta

	(0, 0, 0, 0)		
N_{conf}	(1, 0, 0, 0)	(2, 0, 0, 0)	(3, 0, 0, 0)
	(0, 1, 0, 0)	(0, 2, 0, 0)	(0, 3, 0, 0)
	(0, 0, 1, 0)	(0, 0, 2, 0)	(0, 0, 3, 0)
	(0, 0, 0, 1)	(0, 0, 0, 2)	(0, 0, 0, 3)
	(1, 1, 0, 0)	(2, 1, 0, 0)	...
	(1, 0, 1, 0)	⋮	⋮
	(1, 0, 0, 1)	⋮	⋮
	⋮		



Vibrational WE Methods



- The Hamiltonian matrix grows extremely fast
- Single-shot diagonalization
- High time and memory consumption

Vibrational Configuration Interaction

✓ Ansatz: $\Psi_s(\{Q\}) = \sum^{N_{\text{conf}}} A_{n,s} \Phi^n(\{Q\})$

Example: 4 modes, 3 quanta

	(0, 0, 0, 0)		
	(1, 0, 0, 0)	(2, 0, 0, 0)	(3, 0, 0, 0)
	(0, 1, 0, 0)	(0, 2, 0, 0)	(0, 3, 0, 0)
	(0, 0, 1, 0)	(0, 0, 2, 0)	(0, 0, 3, 0)
	(0, 0, 0, 1)	(0, 0, 0, 2)	(0, 0, 0, 3)
N_{conf}	(1, 1, 0, 0)	(2, 1, 0, 0)	...
	(1, 0, 1, 0)	⋮	⋮
	(1, 0, 0, 1)	⋮	⋮
	⋮		



Evaluation of IR Intensities

$$I_{i \rightarrow f}^{\text{IR}} \propto \sum_{a=x,y,z} \sum_{k=1}^M \left(\frac{\partial \mu_a}{\partial Q_k} \right)_{\text{eq}}^2 \langle i | \hat{Q}_k | f \rangle^2$$



Evaluation of IR Intensities

$$I_{i \rightarrow f}^{\text{IR}} \propto \sum_{a=x,y,z} \sum_{k=1}^M \left(\frac{\partial \mu_a}{\partial Q_k} \right)_{\text{eq}}^2 \langle i | \hat{Q}_k | f \rangle^2$$

where:

$$\left(\frac{\partial \mu_a}{\partial Q_k} \right)_{\text{eq}}$$

are the Born charges computed via CPKS



Evaluation of IR Intensities

$$I_{i \rightarrow f}^{\text{IR}} \propto \sum_{a=x,y,z} \sum_{k=1}^M \left(\frac{\partial \mu_a}{\partial Q_k} \right)_{\text{eq}}^2 \langle i | \hat{Q}_k | f \rangle^2$$

where:

$\left(\frac{\partial \mu_a}{\partial Q_k} \right)_{\text{eq}}$ are the Born charges computed via CPKS

$\langle i | \hat{Q}_k | f \rangle$ are integrals between initial and final states computed analytically



Evaluation of Raman Intensities

$$I_{i \rightarrow f}^{\text{Raman}} = I_{\parallel, i \rightarrow f}^{\text{Raman}} + I_{\perp, i \rightarrow f}^{\text{Raman}}$$



Evaluation of Raman Intensities

$$I_{i \rightarrow f}^{\text{Raman}} = I_{\parallel, i \rightarrow f}^{\text{Raman}} + I_{\perp, i \rightarrow f}^{\text{Raman}}$$

$$I_{\parallel, i \rightarrow f}^{\text{Raman}} \propto \left(\frac{3}{45} B^2 \right)$$

$$B^2 = \frac{1}{2} \left[(P_{xx} - P_{yy})^2 + (P_{yy} - P_{zz})^2 + (P_{zz} - P_{xx})^2 + 6 (P_{xy}^2 + P_{xz}^2 + P_{yz}^2) \right]$$



Evaluation of Raman Intensities

$$I_{i \rightarrow f}^{\text{Raman}} = I_{\parallel, i \rightarrow f}^{\text{Raman}} + I_{\perp, i \rightarrow f}^{\text{Raman}}$$

$$I_{\parallel, i \rightarrow f}^{\text{Raman}} \propto \left(\frac{3}{45} B^2 \right)$$

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$$P_{ab} = \sum_{k=1}^M \left(\frac{\partial \alpha_{ab}}{\partial Q_k} \right)_{\text{eq}} \langle i | \hat{Q}_k | f \rangle$$



Evaluation of Raman Intensities

$$I_{i \rightarrow f}^{\text{Raman}} = I_{\parallel, i \rightarrow f}^{\text{Raman}} + I_{\perp, i \rightarrow f}^{\text{Raman}}$$

$$I_{\parallel, i \rightarrow f}^{\text{Raman}} \propto \left(\frac{3}{45} B^2 \right)$$

$$B^2 = \frac{1}{2} \left[(P_{xx} - P_{yy})^2 + (P_{yy} - P_{zz})^2 + (P_{zz} - P_{xx})^2 + 6 (P_{xy}^2 + P_{xz}^2 + P_{yz}^2) \right]$$

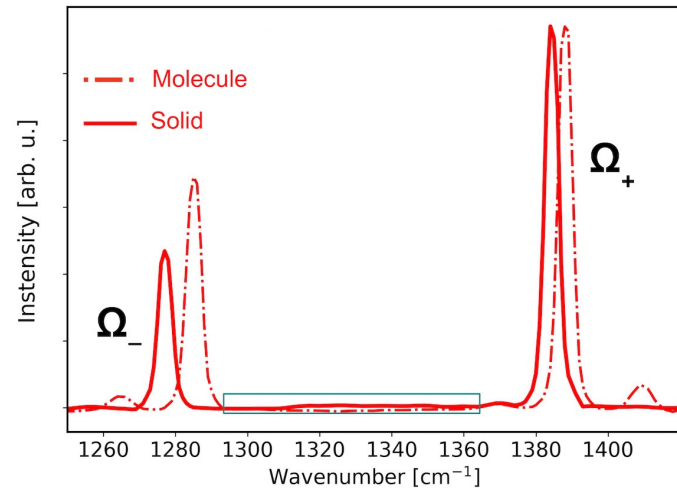
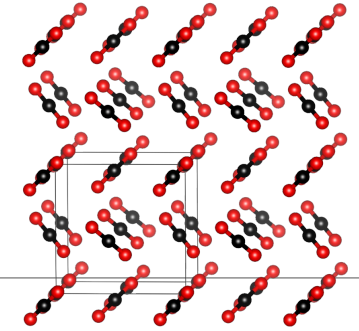
$$P_{ab} = \sum_{k=1}^M \left(\frac{\partial \alpha_{ab}}{\partial Q_k} \right)_{\text{eq}} \langle i | \hat{Q}_k | f \rangle$$

where:

$\left(\frac{\partial \alpha_{ab}}{\partial Q_k} \right)_{\text{eq}}$ are derivatives of the polarizability tensor

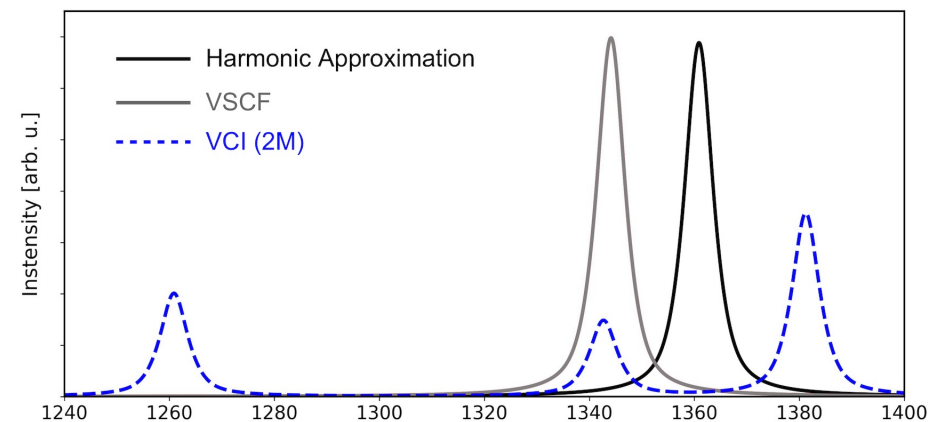
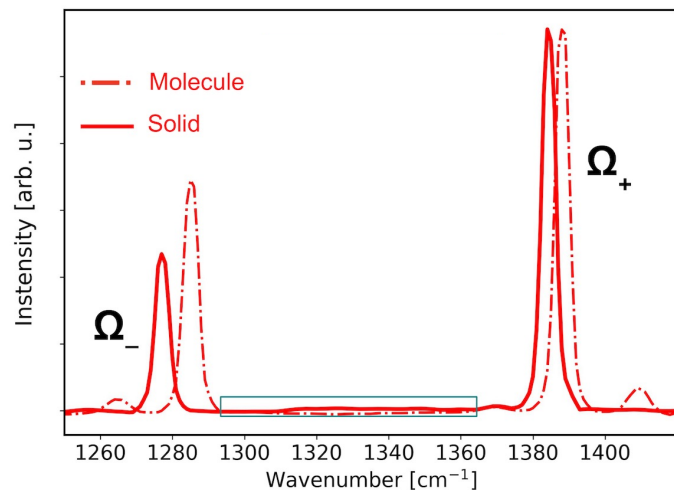
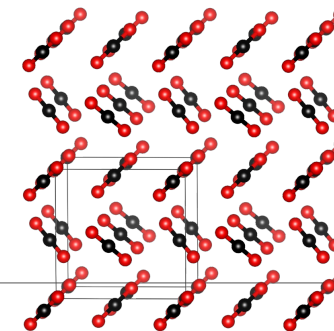


Fermi Resonance in Dry Ice



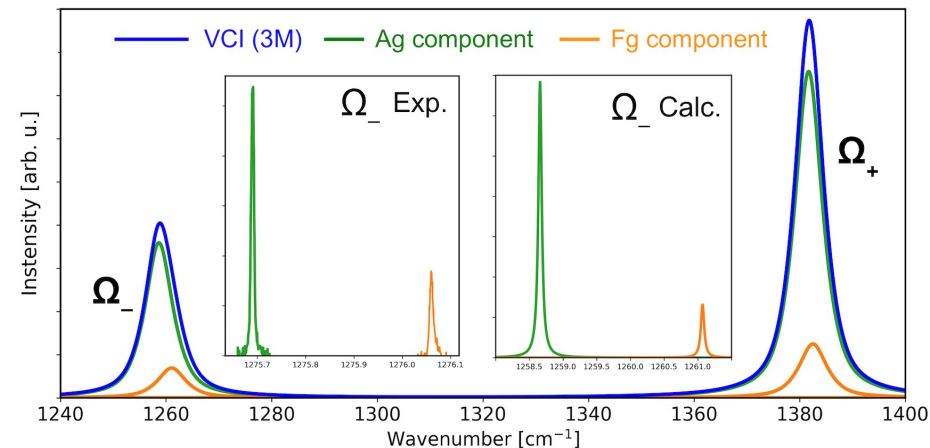
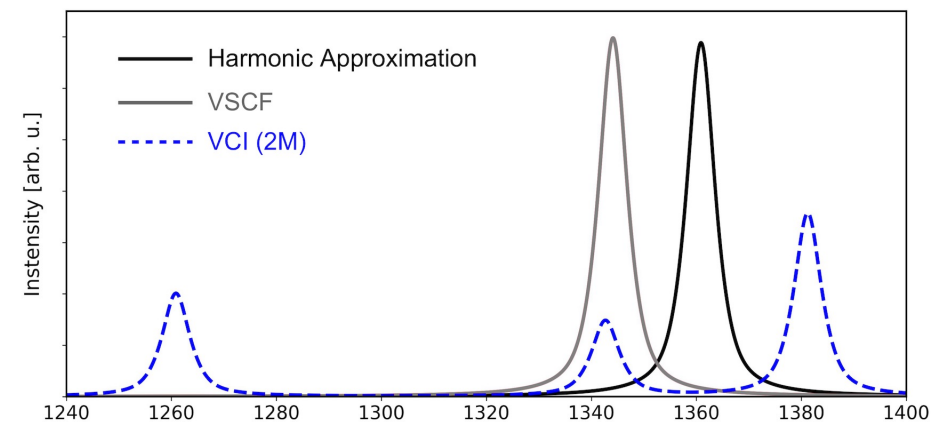
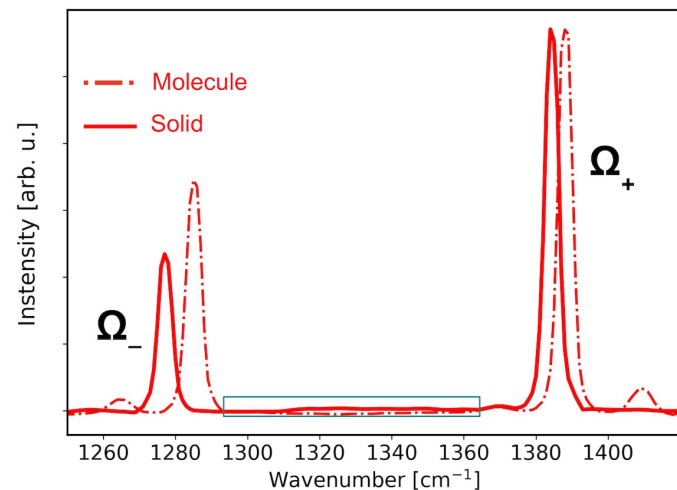
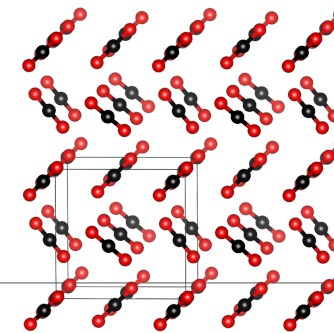


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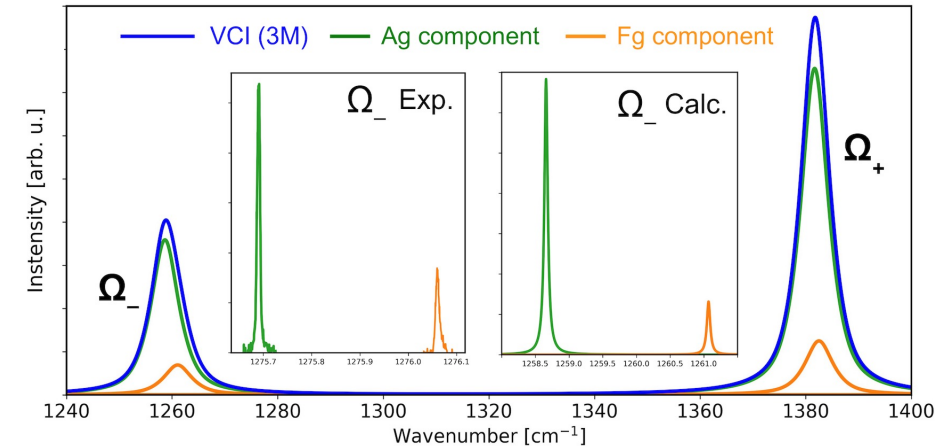
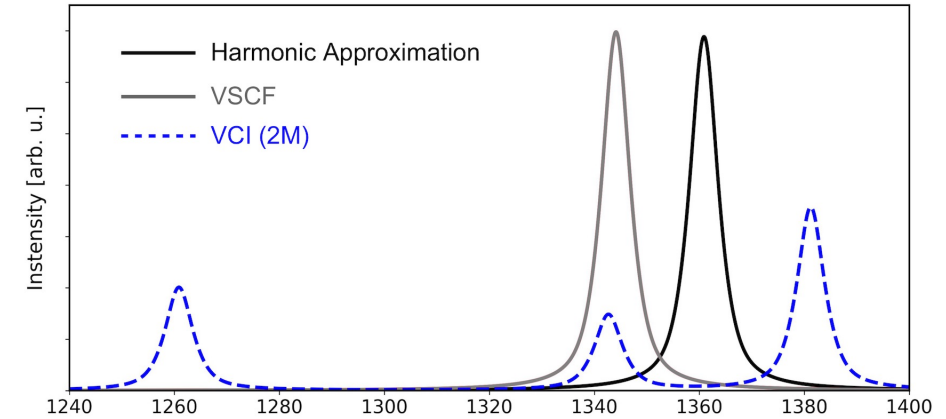
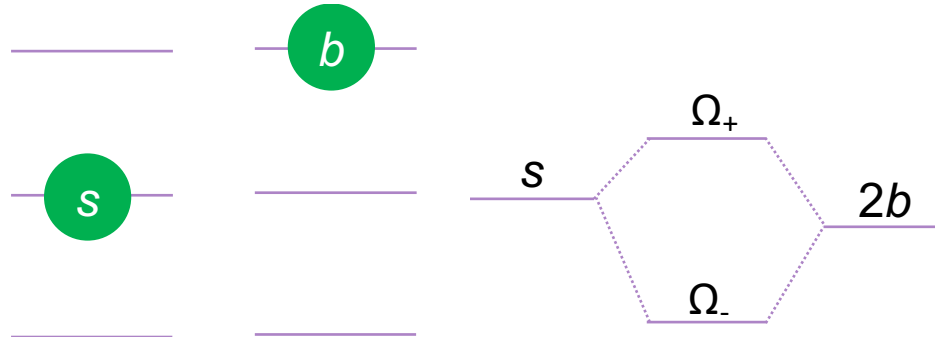
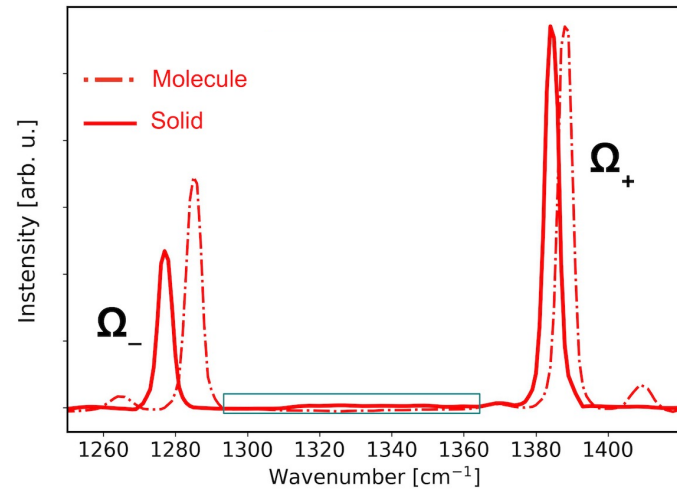
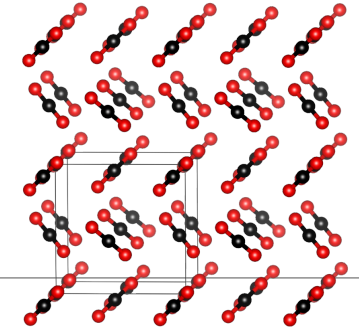


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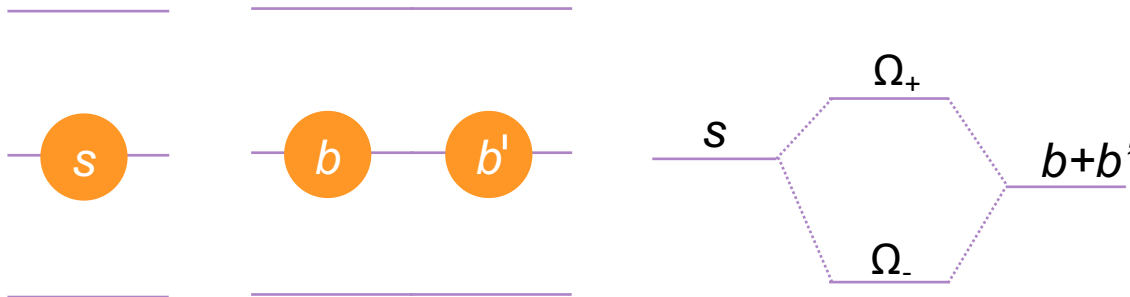
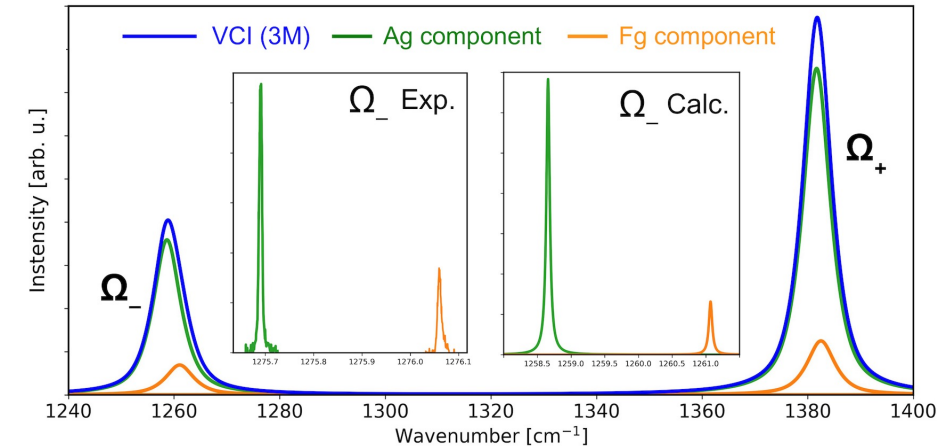
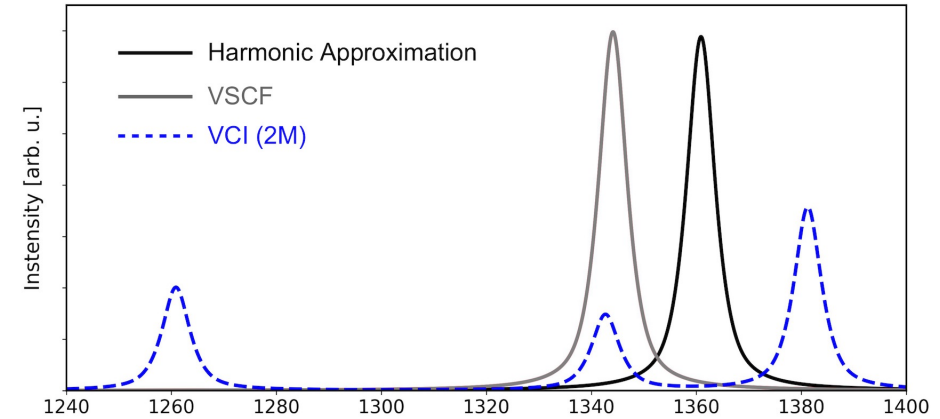
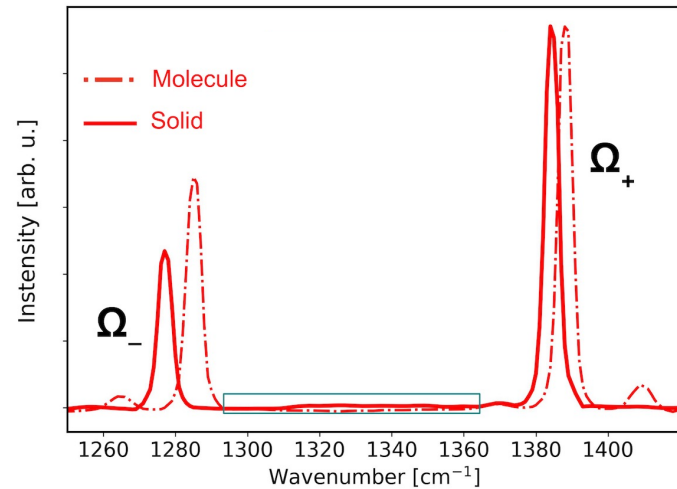
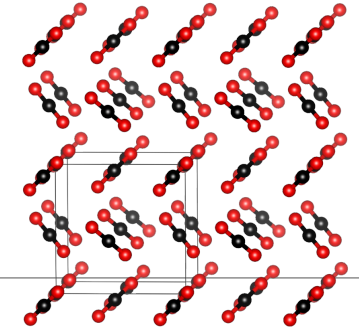


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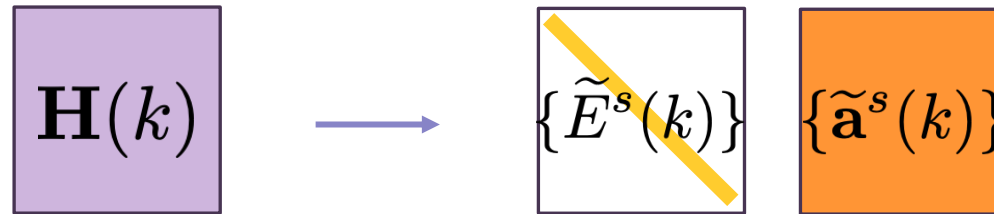


Fermi Resonance in Dry Ice



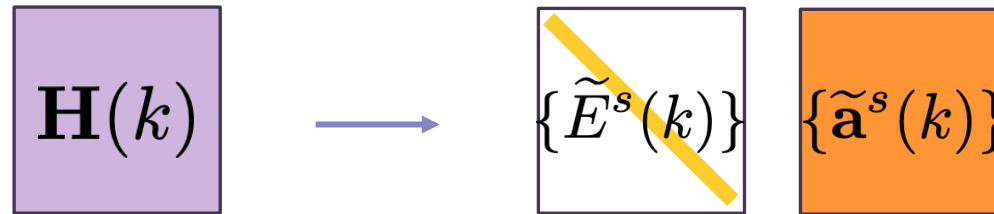
Scaling-Up VCI

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 - ✓ The dimension of the Hamiltonian matrix grows combinatorially with the number of atoms
 - ✓ The rate determining step is the diagonalization procedure (memory and time)



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- ✓ Possible solutions:
 - ✓ “Matrix-free” diagonalization approaches
 - ✓ Forget about VCI $\rightarrow$ perturbative methods

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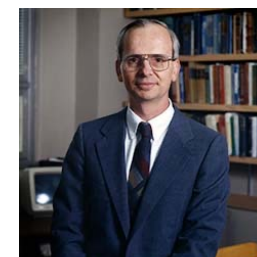
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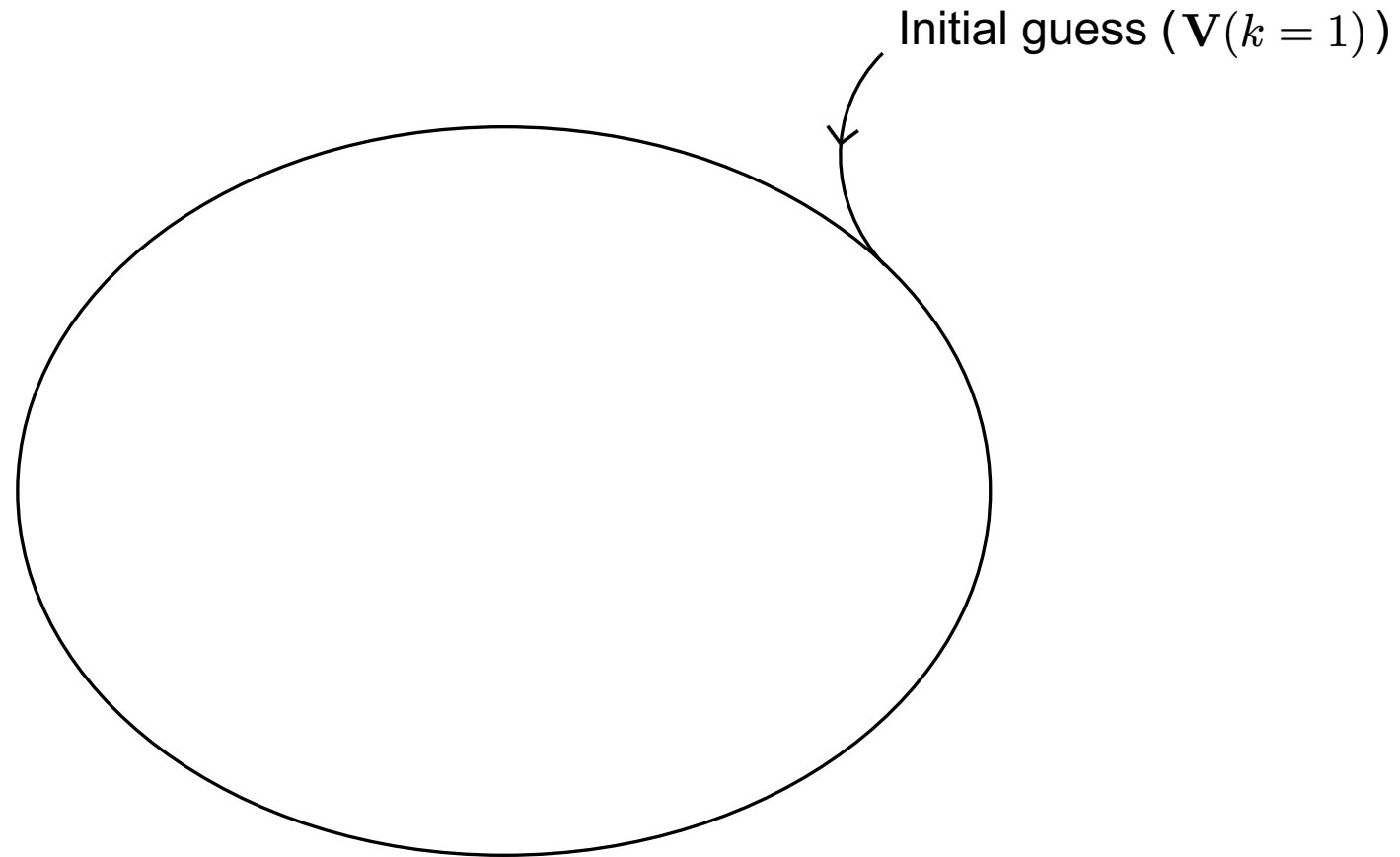


Matrix-Free Diagonalization Methods

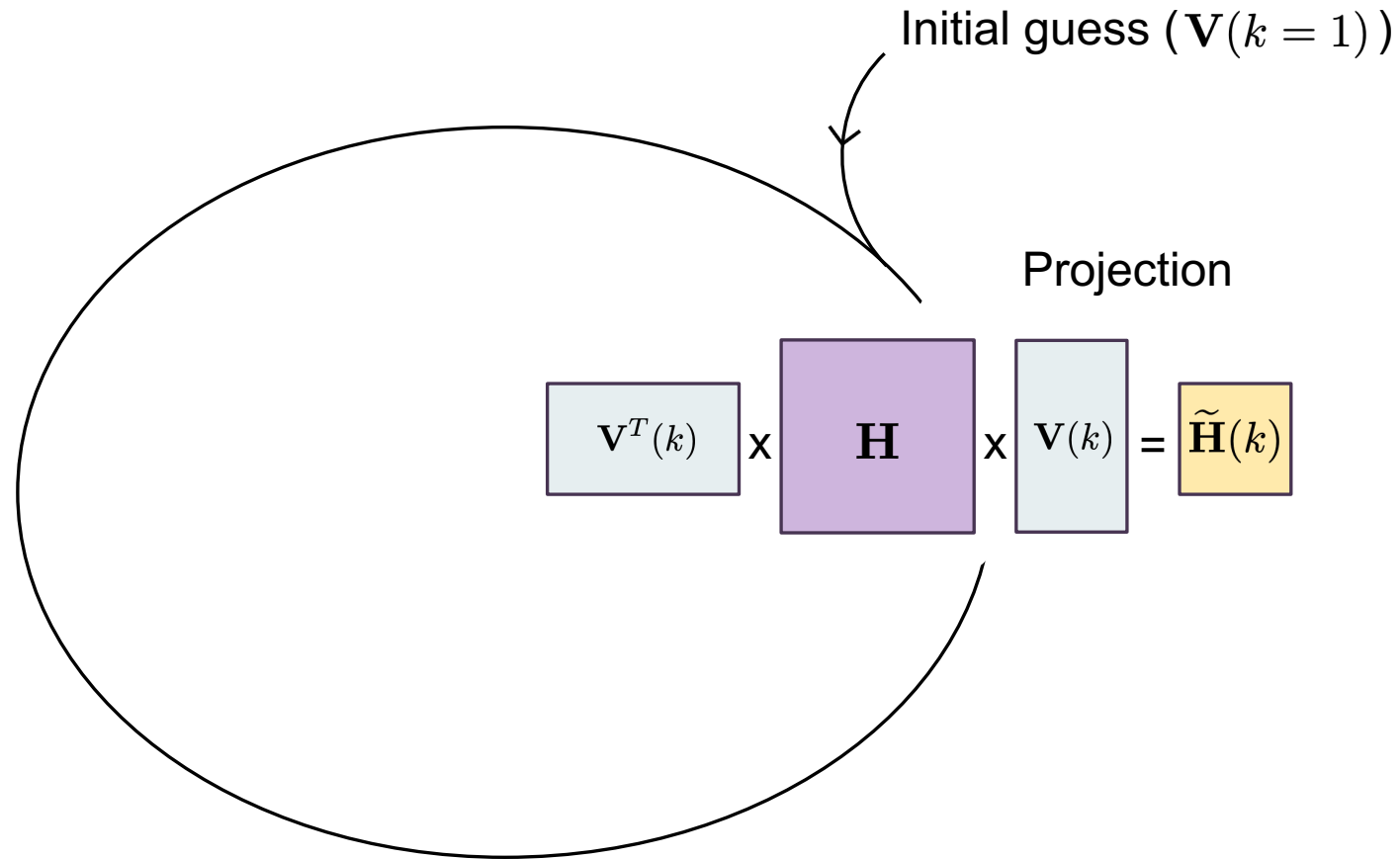
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- ✓ In collaboration with Dr. Jacques Desmarais, J-P’s code was merged in the CRYSTAL source and generalized to VCI



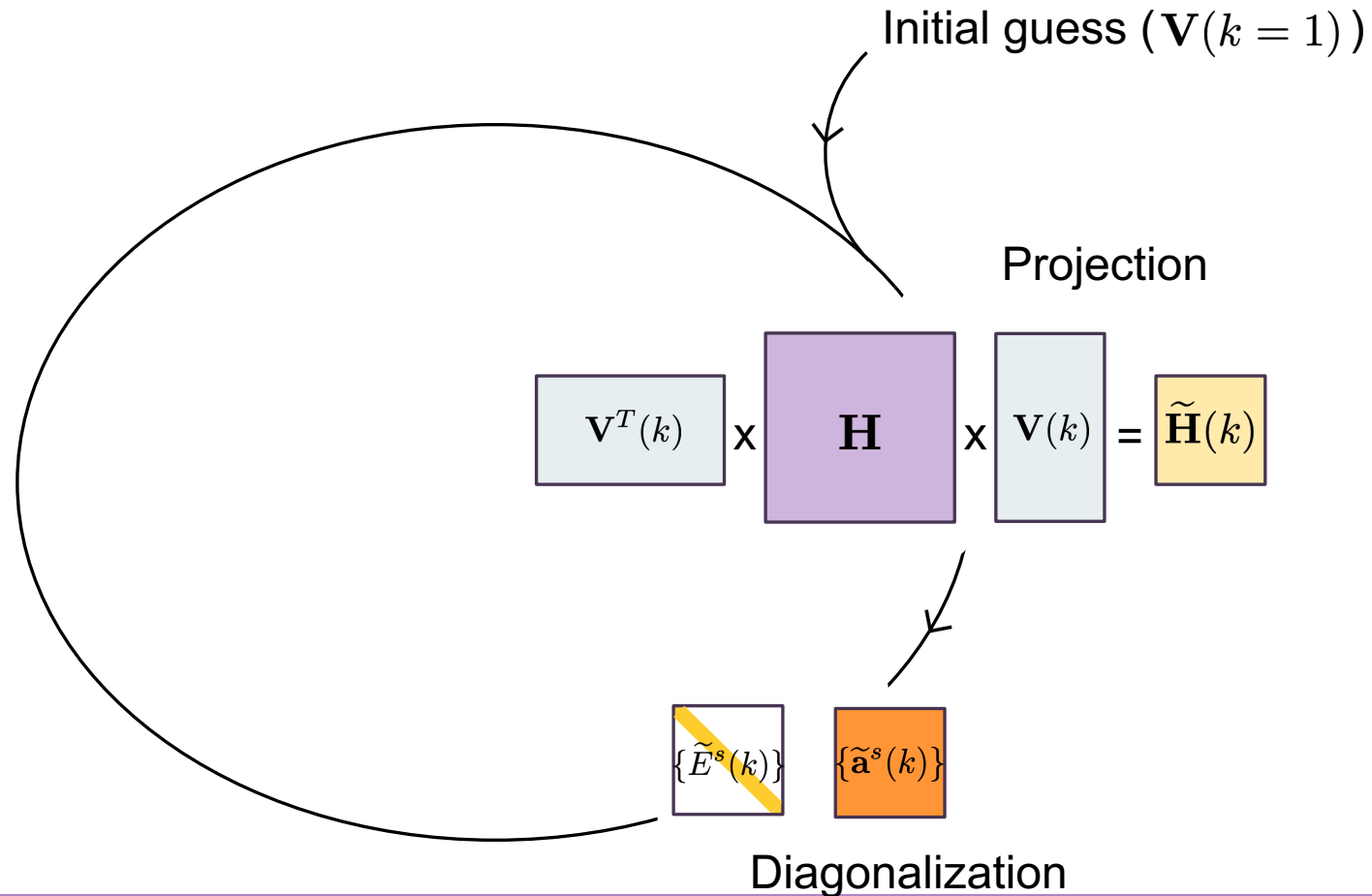
Davidson Method: the Algorithm



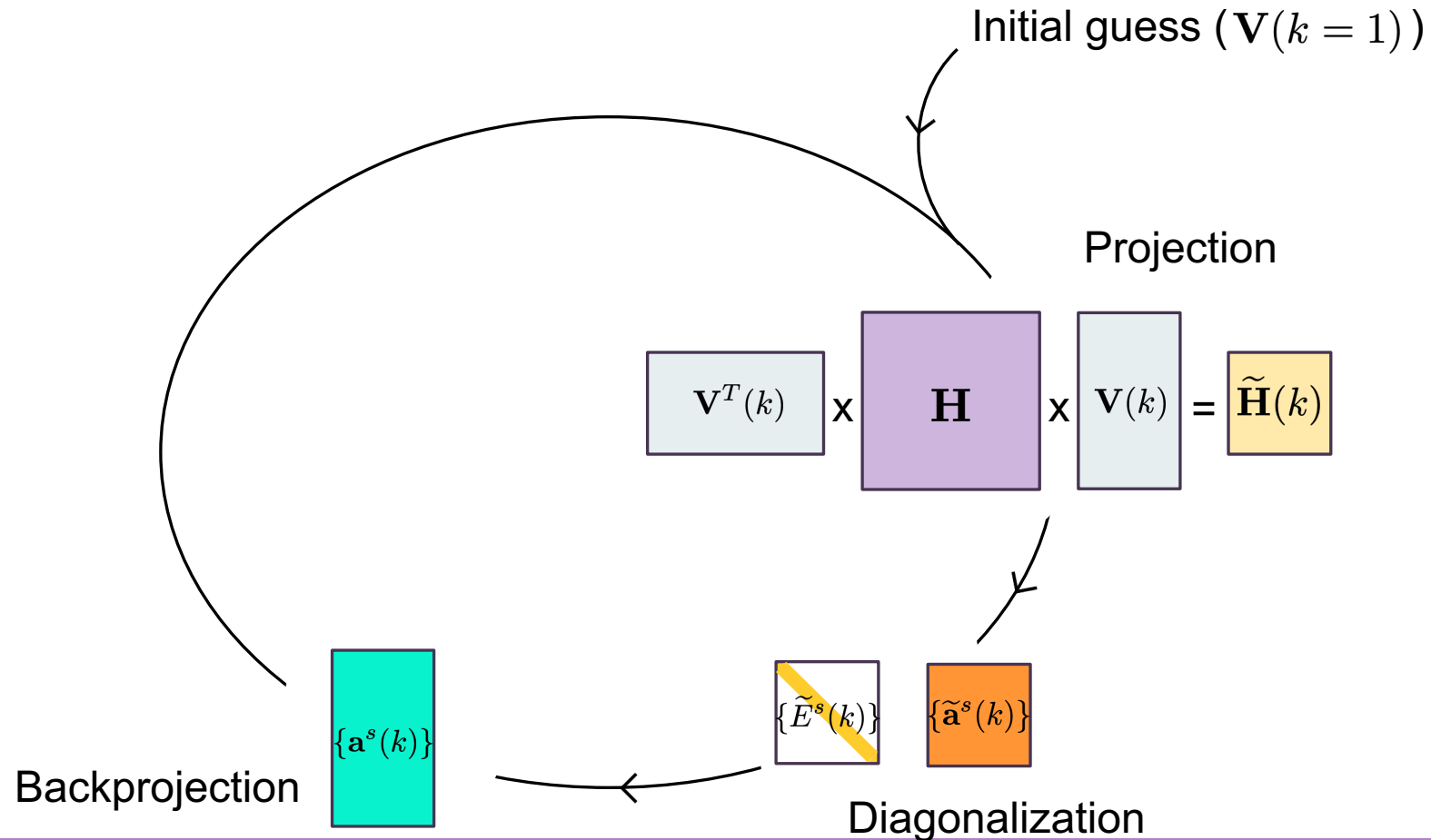
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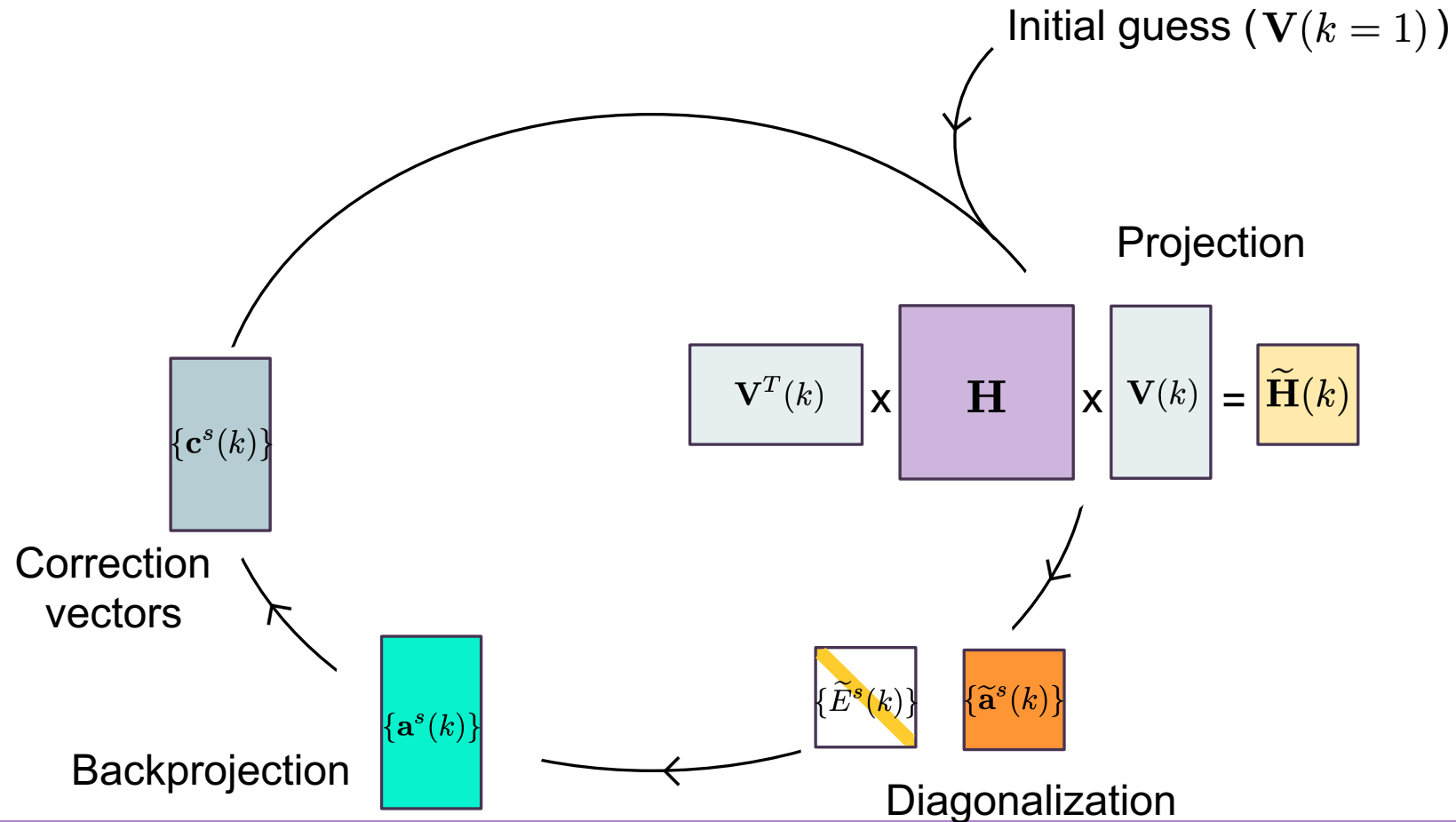
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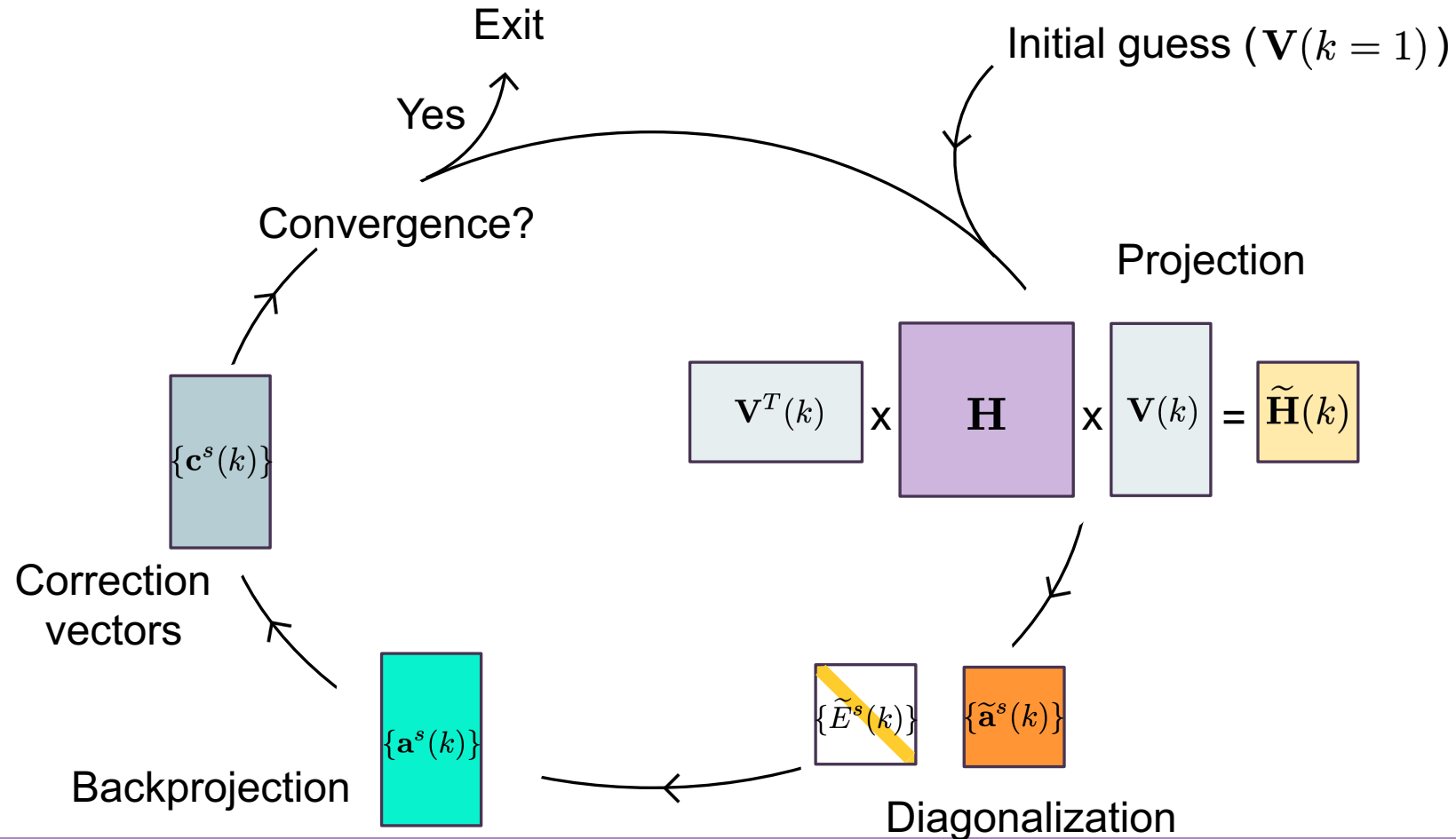
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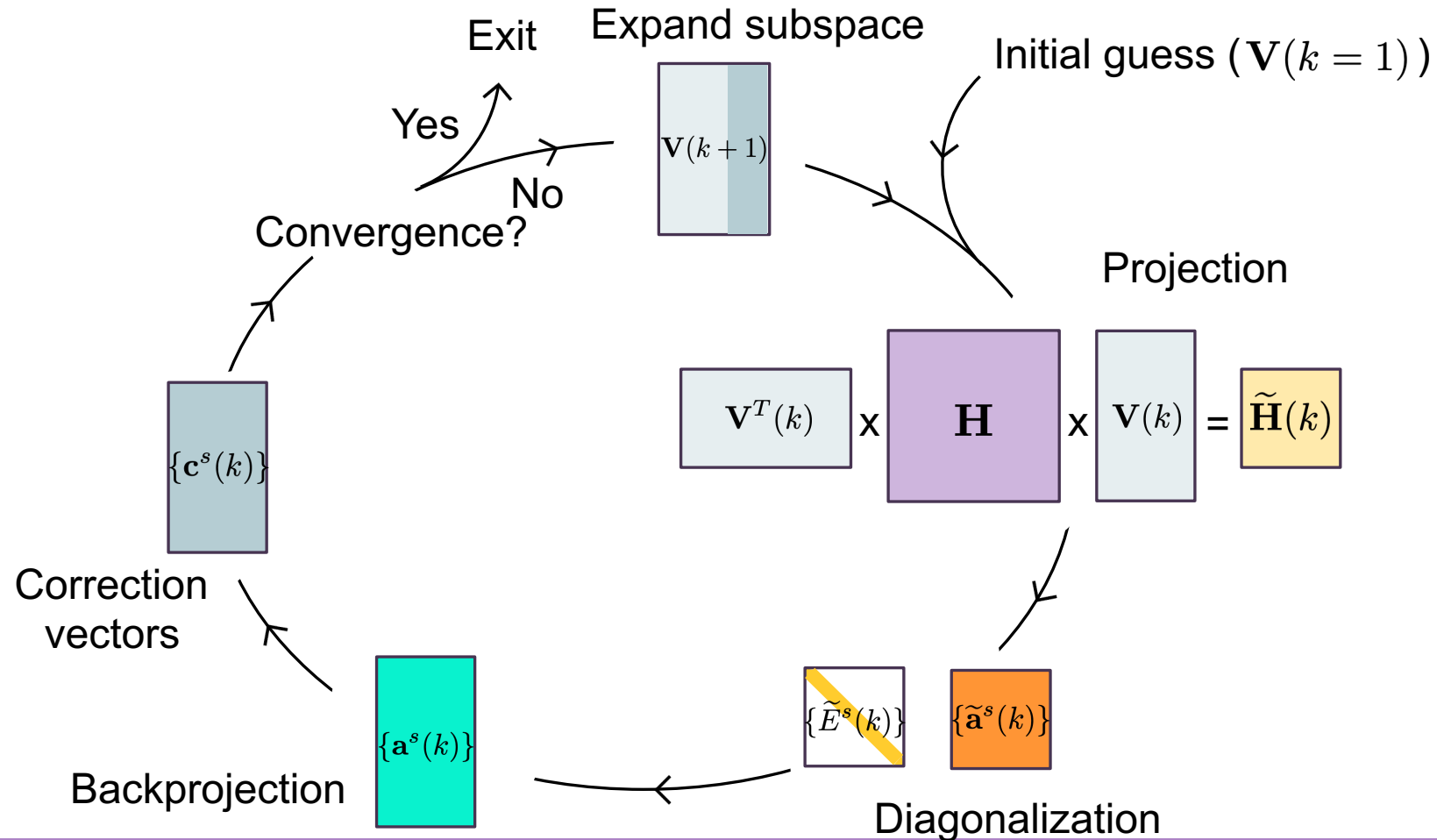
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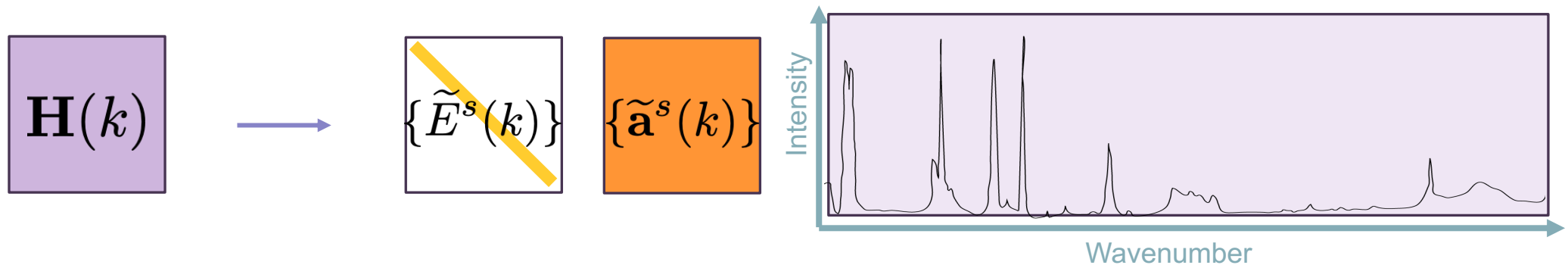


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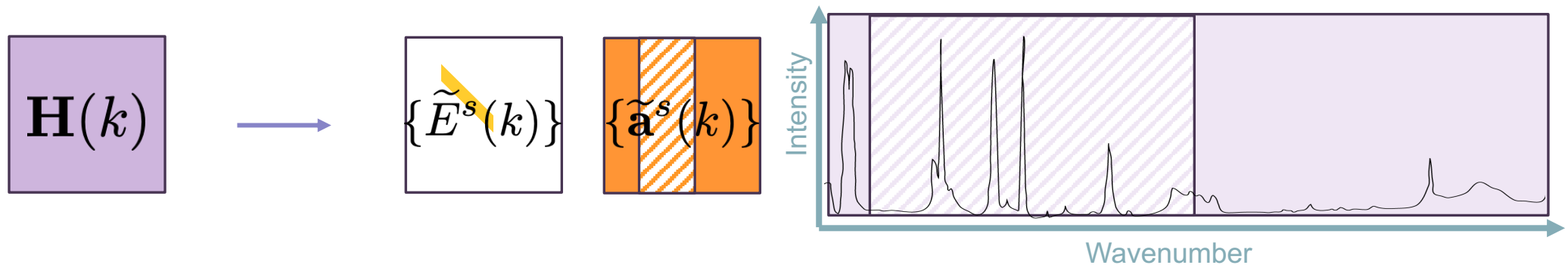
Implementation and Preliminary Results

- ✓ Details of the implementation
 - ✓ The full Hamiltonian is never kept in memory
 - ✓ Matrix multiplications and diagonalizations are handled by BLAS and LAPACK libraries
 - ✓ Several eigenpairs are corrected at each cycle
 - ✓ Eigenpairs are computed for a user-defined energy range



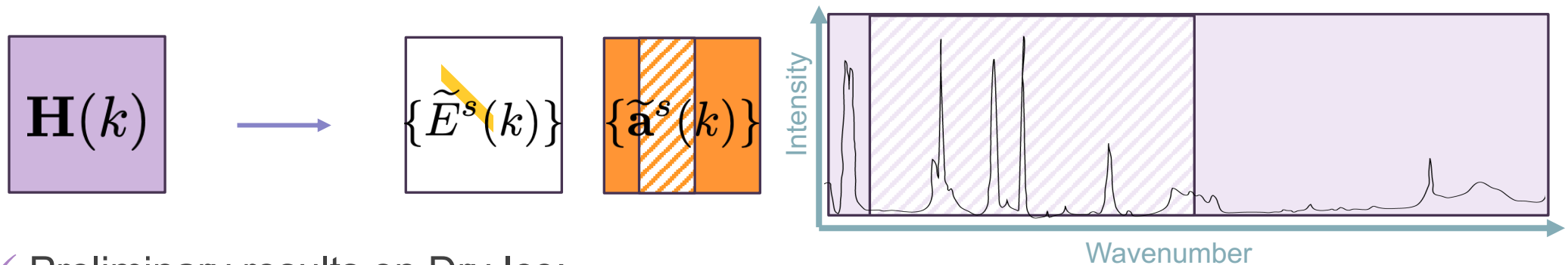
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- ✓ Preliminary results on Dry Ice:
 - ✓ Raman spectrum: 67 s (6 cycles) vs 2410 s. **Davidson is 36 times faster**

Thank You!

More on anharmonicity tomorrow afternoon! (Advanced tutorials)



References:

- ❖ Erba, A., Maul, J., Ferrabone, M., Carbonnière, P., Rérat, M. and Dovesi, R., 2019. Anharmonic vibrational states of solids from DFT calculations. Part I: Description of the potential energy surface. *Journal of chemical theory and computation*, 15(6), pp.3755-3765.
- ❖ Lin, C.Y., Gilbert, A.T. and Gill, P.M., 2008. Calculating molecular vibrational spectra beyond the harmonic approximation. *Theoretical Chemistry Accounts*, 120, pp.23-35.
- ❖ Mitoli, D., Maul, J. and Erba, A., 2023. Anharmonic terms of the potential energy surface: A group theoretical approach. *Crystal Growth & Design*, 23(5), pp.3671-3680.
- ❖ Erba, A., Maul, J., Ferrabone, M., Dovesi, R., Rérat, M. and Carbonnière, P., 2019. Anharmonic vibrational states of solids from DFT calculations. Part II: Implementation of the VSCF and VCI methods. *Journal of Chemical Theory and Computation*, 15(6), pp.3766-3777.
- ❖ Mitoli, D., Maul, J. and Erba, A., 2024. First-Principles Anharmonic Infrared and Raman Vibrational Spectra of Materials: Fermi Resonance in Dry Ice. *The Journal of Physical Chemistry Letters*, 15(4), pp.888-894.